

Selectivity and Multiplicative Gating under Matched-Noise Cross-Condition Shift: Controlled Ablations of a Mamba-3 Block

Yi Wang*

College of Engineering, Lishui University, Lishui 323000, Zhejiang, China

Abstract

The robustness of Mamba-style selective state-space models is often attributed to input-dependent selectivity. We test this attribution with pre-specified ablations of one Mamba-3 block on bearing vibration data under cross-condition shift (XJTU-SY, one speed-and-load pair, training and evaluation bearings disjoint) with matched-noise training, reporting macro-F1 at the final epoch of a fixed budget. The central contrasts were re-run with training and evaluation noise drawn independently. Making the scan’s dynamics and read/write coefficients input-independent (Δ , A , B , C , trapezoidal weight and rotation angles) does not remove the block’s advantage over a time-invariant S4D: the fully frozen block leads by +21.3, +23.3, +20.3 pp at 0, −2 and −6 dB (all intervals excluding zero). Removing the multiplicative output gate costs +9.4, +15.7, +27.0 pp in the partially frozen block and +5.4, +9.3, +32.6 pp in the fully frozen block, resolved only at −6 dB; without its gate the fully frozen block still exceeds S4D at 0 and −2 dB but falls below it at −6 dB. Replacing the gate by an additive branch of identical size and initial weights leaves the result indistinguishable from removing it. Adding a gate to S4D recovers part of the gap, no more than an additive branch of the same size on one transition direction and slightly more on the other. On Paderborn data the gate-removal cost replicates when training and evaluation share bearings, the freeze advantage only at −6 dB, and nothing replicates on disjoint bearings, where every architecture transfers poorly. The evidence supports input-independent scan coefficients sufficing in this block

*Corresponding author

Email address: jefflsxy@lsu.edu.cn (Yi Wang)

and a gate whose necessity grows with noise depth; it is limited to one block, one shift pair per dataset and five seeds.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, matched-noise training

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet recent evidence questions it: a systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds the *simpler, time-invariant* S4D consistently ahead of Mamba-family variants in accuracy and efficiency [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (up to 18 pp), the evaluation labels are identical across all 865 cells of the original grids (hash-verified), and the block can be dissected.

Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. Instead we take a regime in which a Mamba-style block exhibits a large, stable robustness advantage over a time-invariant SSM, and ask which ingredient earns it. The chain has three steps, each with adjudication thresholds fixed before the corresponding grid was run:

1. **Test the classical selectivity terms (necessity test).** We construct a *frozen* arm: the input-dependent parts of Δ , A , B and C are replaced by learned constants. The Mamba-3 scan also takes an input-dependent trapezoidal weight and input-dependent rotation angles; these, the convolutional stem, the multiplicative SiLU gate, the s-

tate parameterisation, normalisation and projections are retained (Section 3.2). If these four terms drive robustness, the frozen arm should collapse toward the time-invariant baseline.

2. **Localise the necessary component.** On the frozen base, we remove one component at a time (convolutional stem; SiLU gate; heavy-tailed state parameterisation) under a single-variable discipline checked key-by-key on the state dictionary (Section 3.2), and adjudicate each arm against pre-specified Δ -thresholds.
3. **Test sufficiency by constructive grafting.** The component identified as necessary is added to plain S4D as a gate branch, and a pre-specified recovery ratio decides whether “removal collapses” is matched by “addition restores”.

The outcome is a conditional pattern. Under independent training and evaluation noise (Section 4.1), making the scan’s dynamics and read/write coefficients input-independent does not remove the block’s advantage over S4D (+21.3, +23.3, +20.3 pp for the fully frozen block). Removing the gate costs +9.4, +15.7, +27.0 pp in the partially frozen block, resolved only at −6 dB, and without its gate the fully frozen block still exceeds S4D at 0 and −2 dB but not at −6 dB: the gate’s contribution grows with noise depth. Replacing the gate by an additive branch of identical size cannot be distinguished from removing it, in either host. Adding a gate to S4D, rebuilt in the donor’s width, recovers $R = 0.30$ of the gap (paired-seed interval 0.14–0.72), not more than an additive branch with identical parameters (0.41) on the original transition, and somewhat more on the reverse one (Sections 4.6 and 4.8). On Paderborn the gate-removal cost replicates when training and evaluation share bearings and nothing replicates on disjoint bearings, where no architecture transfers well (Section 4.9). A per-class analysis on a separate, shorter grid indicates that the gate-related differences concentrate on the minority class.

Contributions. (i) A layered freezing experiment on a Mamba-3 block under operating-condition shift and matched noise, from the classical selectivity terms (Δ , A , B , C) to a scan whose dynamics and read/write coefficients are all input-independent, in which the block’s deep-noise advantage over S4D survives every step, including under independent training and evaluation noise; each arm’s frozen coefficients are verified input-independent at the kernel interface. (ii) Same-host and cross-host tests of the output gate: removal, in-place replacement by an additive branch of identical size,

68 and grafting onto S4D with an additive control, showing a gate contribution
69 that is resolved at the deepest noise level and an additive branch that does
70 not substitute for it, while a time-invariant host gains little that is specific
71 to gating. (iii) A correction of the noise protocol (training and evaluation
72 noise originally shared by index) with a pre-specified re-run, and final-epoch
73 reporting under a fixed budget with best-epoch selection as a sensitivity
74 analysis. (iv) A documented pre-specification and review trail, including the
75 incidents in which automated adjudication departed from the pre-specified
76 criteria; we release the trail.

77 2. Background and Related Work

78 2.1. Time-invariant and selective SSMs

79 Diagonal structured SSMs (S4D) [6] evolve a linear state with input-
80 independent parameters (A , B , C , Δ), acting as a bank of learned linear
81 filters. Mamba-style blocks [1] make Δ , B , C functions of the input token
82 (selectivity), wrap the scan in a block containing a short causal convolu-
83 tional stem, a multiplicative SiLU gate on a parallel branch, and residu-
84 al/normalisation structure; the concrete block used throughout this paper
85 follows the Mamba-3 formulation [7]. The narrative crediting selectivity for
86 robustness draws on theory showing input-dependent dynamics can represent
87 discontinuity-sensitive projections and counteract memory decay [8], and on
88 [9]. We treat these as *hypotheses to be tested*, not established attribution-
89 s: the theoretical results concern representability, which does not by itself
90 determine which component earns empirical robustness in a trained block.

91 2.2. The unresolved comparison

92 Empirically the picture is contested. A systematic comparison on 59 clean
93 multivariate time-series classification benchmarks (MONSTER and UEA)
94 finds S4D-based models superior to Mamba variants [4]; that study does not
95 include additive noise or a shift of operating condition between training and
96 evaluation, which is the regime examined here; theoretical work identifies
97 long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, appli-
98 cation literatures—including machine-condition monitoring—contain many
99 Mamba-hybrid methods reporting noise robustness [10, 11, 12, 13, 14], but
100 none, to our knowledge, isolates *which component* of the block is responsible
101 under controlled conditions. Our clean-data results agree with the S4D-
102 superiority finding (S4D matches or slightly exceeds the Mamba-style block

on clean data). Whether the disagreement in the literature reflects a regime difference is not settled by a single testbed; what we can report is that, in the drift $\times$ noise regime studied here, the Mamba-style block’s advantage over S4D is not removed by freezing the classical selectivity terms.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-condition drift (speed and load both change) as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (a change of rotation speed and radial load between recorded conditions), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts selective SSMs should excel at [8], which makes it a plausible arena for the selectivity hypothesis; and (c) the robustness gap between the Mamba-style block and S4D is large (up to 20 pp), giving the attribution chain high signal. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [15] are split by operating condition: models train on one operating condition (37.5 Hz, 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change (cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Section 4.11). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Section 4.11 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every cell on the original transition carries `eval_sha = 6c20b367...` (Appendix A), recomputed independently during adjudication; the reverse transition and the two Paderborn settings carry their own label hashes (93424834..., 53d565d9... disjoint, addc1938... shared). The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75), and follow-up grids add 585 cells (Sections 4.1, 4.3, 4.10, 4.6, 4.7, 4.8 and 4.9).

138 Training and evaluation bearings are disjoint in each direction: Bearing2_1,
 139 2_2, 2_4, 2_5 at 37.5 Hz/11 kN and Bearing3_1, 3_3, 3_4, 3_5 at 40 Hz/10 kN;
 140 the reverse transition swaps their roles.

141 **Composition of the data.** Training uses the bearings of the 37.5 Hz/11 kN
 142 condition whose failure mode is inner race or outer race (Bearing2_1 is la-
 143 belled IR; Bearing2_2, 2_4 and 2_5 are labelled OR); evaluation uses those of
 144 the 40 Hz/10 kN condition (Bearing3_3 and 3_4 are IR; Bearing3_1 and 3_5
 145 are OR). Bearings with cage or mixed failure modes are excluded. Windows
 146 are non-overlapping segments of 2048 samples of the horizontal channel taken
 147 from the fault phase, which is delimited by a kurtosis/RMS onset criterion
 148 with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR
 149 windows, the training set 4,624 OR and 624 IR windows. Two consequences
 150 matter for interpretation. The IR class is learned from a single training bear-
 151 ing, so class-level results may reflect bearing-specific generalisation. And the
 152 five seeds resample initialisation, batch order and noise but not bearings:
 153 there are four independent bearings on each side, and none of the intervals
 154 reported here captures bearing-level uncertainty. One follow-up grid (Sec-
 155 tion 4.8) reverses the transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). It uses the
 156 same eight bearings with their roles swapped, so it tests the other direction
 157 rather than independent bearings; its evaluation set holds 4,624 OR and
 158 624 IR windows, all IR windows coming from Bearing2_1, and its label hash
 159 is 93424834.... The 35 Hz/12 kN condition has no inner-race bearing and
 160 cannot form a binary pair.

161 The principal arms are summarised in Table 1.

162 3.2. *What the frozen arm freezes, and what the checks verify*

163 In the Mamba-3 layer used here, one input projection emits the gate and
 164 value streams, the scan inputs B and C , the step-size and decay terms (Δ
 165 and A , through input-dependent additive terms), a trapezoidal weight and
 166 rotation angles. The frozen arm removes the input-dependent parts of Δ ,
 167 A , B and C and replaces them with learned constants broadcast over time
 168 (for Δ the static bias of the original layer is kept; for A a new constant pa-
 169 rameter plays the role of a static log-decay, passed through the layer’s own
 170 heavy-tailed clamp). The removed projections are deleted, so no parameter
 171 or gradient can leak. **The trapezoidal weight and the rotation angles**
 172 **remain functions of the current input**, as do the stem and the output
 173 gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3
 174 formulation [7] the input-dependent rotation and trapezoidal weighting also

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.4)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.4)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.4)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: scan dynamics and read/write coefficients input-independent (value stream and gate still input-dependent)	103,842
frozen-all-gate	frozen-all with the gate rows removed from the input projection (Section 4.1)	71,074
frozen-all+add	frozen-all with $y \odot \text{SiLU}(z)$ replaced by $y + \text{SiLU}(z)$; same parameters and initial weights	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.5)	116,802
S4D+gate (wm)	Graft rebuilt in the donor’s shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\text{out}}(y \odot \text{SiLU}(z))$ (Section 4.6)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\text{out}}(y + \text{SiLU}(z))$ (control)	198,978
frozen+add	BM3-frozen with the gate replaced in place: $W_{\text{out}}(y + \text{SiLU}(z))$ instead of $W_{\text{out}}(y \odot \text{SiLU}(z))$; same modules, parameters and initial weights	112,410

175 enter the state update. Section 4.2 is accordingly a statement about the
 176 classical selectivity terms Δ, A, B, C only. Three further arms close that gap
 177 by the same device: *frozen-trap* replaces the trapezoidal weight by a learned
 178 per-head constant, *frozen-rot* replaces the rotation angles by learned con-
 179 stants (so the rotation becomes a learned position-only phase rather than a
 180 function of the current input), and *frozen-all* does both, so that the scan’s
 181 dynamics and read/write coefficients (Δ, A, B, C , trapezoidal weight, rota-
 182 tion angles) no longer depend on the input. The value stream x and the gate
 183 input z are still computed from the input, as they must be for the layer to
 184 carry any signal; the rotation becomes a learned position-dependent phase.
 185 In each case the corresponding `in.proj` output slice is deleted. Section 4.3
 186 reports them. For these arms the construction check adds the decisive probe:
 187 the tensors handed to the scan are captured for two different inputs through
 188 the same module, and a tensor the arm claims to have frozen is required to
 189 be bitwise identical across them while a tensor left selective is required to
 190 differ (all three arms pass). No kernel modification is needed, because the
 191 scan interface accepts these tensors regardless of their origin.

192 Single-variable discipline was checked by key-level state-dictionary diffs
 193 between two models built under the same seed. For the frozen arm the
 194 diff verifies that every non-target tensor has the same name and shape as in
 195 BM3; for the component-ablation and graft arms it additionally compares the
 196 values of non-target tensors at initialisation. Every differing key is mapped
 197 to an entry in an architecture map. We do not claim byte-level identity of
 198 non-target tensors in the frozen arm, and trained weights naturally differ
 199 between arms; the checks concern how the arms were constructed.

200 3.3. Reporting rule

201 Every cell is trained for a fixed budget of 50 epochs with a cosine learning-
 202 rate schedule that anneals to zero, and the macro-F1 reported in the main
 203 text is that of the *final epoch*, a rule that uses no evaluation data to choose
 204 a model. The original grids recorded instead the maximum over epochs of
 205 macro-F1 on the evaluation split (the harness defines no separate validation
 206 split), and the thresholds pre-specified for those grids were written against
 207 that rule. Every arm of the chain was therefore re-run on the adjudication
 208 band (0, -2, -6 dB) with the original training code, the same evaluation
 209 fingerprint and per-epoch evaluation (Section 4.10). Adjudications are re-
 210 ported under the rule they were specified for and under the final-epoch rule,
 211 and best-epoch values are given as a sensitivity analysis. Levels outside the

band (clean, +10, +6, −10 dB) and the per-class grid were not re-run; values there are best-epoch and labelled auxiliary. Because the final epoch of the same evaluation condition has been inspected repeatedly during this study, final-epoch values are not a fresh confirmatory test either.

3.4. Confound elimination

Before component attribution, three candidate confounds were examined on the same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D (S4D-wide − S4D: −0.4, −1.0, +0.4 pp at 0, −2, −6 dB under final-epoch reporting, all intervals including zero), while BM3-frozen exceeds S4D-wide by +15.9, +20.8, +16.0 pp with intervals excluding zero at all three levels, so this particular increase in S4D’s parameter count does not explain the gap; the control does not address the change in channel mixing that gate removal also entails. **Normalisation:** both SSM arms use LayerNorm and contain no BatchNorm; the arm that degrades under noise is the one *without* any batch statistics, which argues against BatchNorm statistics mismatch as the explanation. Other normalisation inside the Mamba-3 layer was not varied. **Noise streams:** in the original harness the noise added to window i was drawn from a generator keyed on (seed, i), and training and evaluation used the same seed, so window i of the training set and window i of the evaluation set carried the same base noise vector. Per-index label agreement between the two sets is 0.549 on XJTU-SY, below the 0.628 expected by chance, so a model that memorised training noise would be pushed *away* from the correct test label; on Paderborn it is 0.999 (disjoint) and 1.000 (shared). All central contrasts were therefore re-run with the stream keyed on (seed, split, i), which makes training and evaluation noise independent while all arms with a given seed share one evaluation noise bank (Section 4.1). **Auxiliary physics loss:** an exploratory kinematics-alignment regulariser carried by the original model contributes −0.1 pp pointwise (curves coincide); all attribution arms are therefore built on the pure cross-entropy variant.

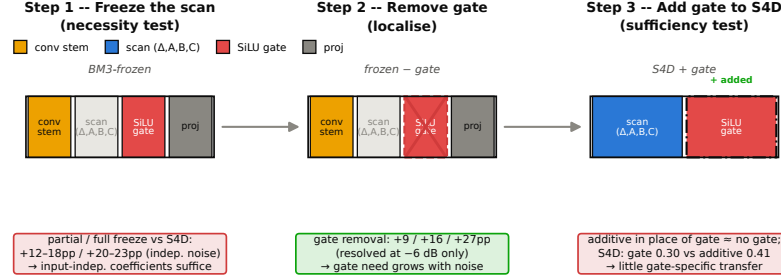
3.5. Pre-specification and adjudication

Every adjudicated question in this study had numeric decision thresholds written to a task card and to disk before the corresponding grid was launched. This is internal pre-specification, not external registration: the supporting evidence is local file modification times and run-directory names (released, [Appendix A](#)), which are not tamper-evident and are not preserved by a repository checkout, so the trail lets readers inspect the order of events but

cannot prove it. The exception is the last three follow-up grids (Sections 4.6–4.10): their thresholds, and for the last two grids the title decision that depends on their outcome, were committed to the public repository before the first cell was run, which gives those grids an external time-stamp. Later questions (component ablation, graft) were specified after earlier grids had been seen, as is natural in a staged design. Adjudication was mechanical: an analysis pass applies the thresholds and emits a decision artifact; an independent reviewer pass re-computes aggregates, verifies fingerprints row-by-row, and must cite file paths and modification times for every check. The pre-specified thresholds were binding in practice: two attempted post-hoc criterion substitutions and one stale-artifact false verification were caught and re-adjudicated under the original rules — this is the one place where pre-specification actually bit. Appendix C details the incidents, the automated executor/reviewer implementation, and the human adjudication gates; all scientific decisions — thresholds, rulings, and interpretations — were made by the authors and are timestamped. Statistical tests (paired, per-condition) are reported throughout but were never adjudication criteria. A single repeat of one 2-epoch configuration in a separate process differed by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions; Appendix B); this is an observed repeat difference, not an estimated variability bound, and arm-level contrasts are judged against seed-level variability (2–9 pp).

4. Results

Throughout, “deep noise” denotes the pre-specified adjudication band $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-invariant baseline is near chance there: a constant majority-class predictor scores 40.0% and a uniformly random predictor 48.6% macro-F1 on this evaluation set, against 52.3% for S4D at the best epoch). Section 4.1 reports the central contrasts under independent training and evaluation noise and is primary for the arms it contains; the later subsections report the full chain under the earlier noise protocol (training and evaluation noise shared by index, Section 3.4), labelled as such. All values are macro-F1 (%), mean over 5 seeds, at the final epoch of the 50-epoch budget (Section 3.3); best-epoch values are given alongside or in Section 4.10, and values outside the adjudication band are best-epoch and auxiliary. Figure 1 summarises the three-step ablation chain that structures this section.



Conditional on this block, testbed and budget: input-independent scan coefficients suffice; the gate matters most in the deepest noise.

Figure 1: The three-step ablation chain with pre-specified thresholds: freeze the input-dependent Δ, A, B, C projections (Section 4.2), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.4), and graft that ingredient onto S4D to test sufficiency (Section 4.5).

4.1. Central contrasts under independent training and evaluation noise

A pre-specified grid of 150 cells (thresholds, labels and writing consequences committed to the public repository before the first cell) re-ran the central arms with independent training and evaluation noise (Section 3.4); everything else is unchanged. Each contrast is labelled *holds* if at least two of the three levels have a positive mean with an interval excluding zero and none is reversed, *weakened* if exactly one, *does not hold* if none. Across the eight re-run arms, the arm means moved by -3.4 to $+3.9$ pp relative to the earlier protocol (Table 2); no contrast changed direction.

Freezing. Both freeze contrasts hold. The fully frozen block exceeds S4D by $+21.3$, $+23.3$, $+20.3$ pp, in every seed at every level, with the smallest seed-to-seed spread of any arm; the partially frozen block exceeds it by $+12.1$, $+18.1$, $+18.4$ pp.

Gate removal. Removing the gate from the partially frozen block costs $+9.4$, $+15.7$, $+27.0$ pp; the point estimates are large and positive in 13 of 15 seed-level pairs, but only the -6 dB interval excludes zero, so under the pre-specified rule the contrast is *weakened* (it held under the earlier protocol). The same test inside the fully frozen block gives $+5.4$, $+9.3$, $+32.6$ pp, again resolved only at -6 dB. Without its gate, the fully frozen block still exceeds S4D by $+15.9$ and $+14.0$ pp at 0 and -2 dB (intervals excluding zero) but falls 12.3 pp below it at -6 dB (interval $[-17.5, -7.0]$, all seeds). At 0 and -2 dB the fully frozen block’s advantage is therefore not carried by the gate; at

Table 2: Independent training and evaluation noise, original XJTU-SY transition. Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds. Lower block: paired differences with 95% t -intervals and the number of seeds (of 5) with a positive difference, and the pre-specified label. frozen-all-gate and frozen+add variants are the same-host tests of the gate in the fully frozen block.

Arm / contrast	0 dB	−2 dB	−6 dB	label
BM3-frozen	80.3	86.4	87.0	
frozen-all	89.5	91.5	88.9	
S4D	68.3	68.3	68.6	
frozen-gate	70.9	70.7	60.0	
frozen+add	71.3	71.3	60.1	
frozen-all-gate	84.1	82.2	56.3	
frozen-all+add	84.9	83.6	65.6	
frozen-all	+21.3 [14.6, 27.9]	+23.3 [20.2, 26.4]	+20.3 [17.9, 22.8]	
– S4D	(5/5)	(5/5)	(5/5)	holds
BM3-frozen	+12.1 [0.9, 23.2]	+18.1 [9.8, 26.4]	+18.4 [12.7, 24.0]	
– S4D	(5/5)	(5/5)	(5/5)	holds
BM3-frozen	+9.4 [−0.5, 19.4]	+15.7 [−0.8, 32.1]	+27.0 [7.0, 46.9]	
– frozen-gate	(5/5)	(4/5)	(4/5)	weakened
frozen-all	+5.4 [−3.8, 14.6]	+9.3 [−0.5, 19.1]	+32.6 [28.9, 36.3]	
– frozen-all-gate	(3/5)	(5/5)	(5/5)	weakened
frozen-all-gate	+15.9 [6.9, 24.8]	+14.0 [2.5, 25.4]	−12.3 [−17.5, −7.0]	
– S4D	(5/5)	(4/5)	(0/5)	descriptive
frozen+add	+0.4 [−0.6, 1.4]	+0.6 [−0.1, 1.2]	+0.1 [−0.4, 0.7]	
– frozen-gate	(4/5)	(5/5)	(3/5)	descriptive
frozen-all+add	+0.8 [−1.5, 3.1]	+1.4 [−1.9, 4.7]	+9.3 [−7.2, 25.8]	
– frozen-all-gate	(2/5)	(2/5)	(5/5)	descriptive

305 −6 dB the gate is needed for it. The pre-specified condition for a combined
 306 attribution (the fully frozen block’s advantage depends on the gate: gate
 307 removal holds and the additive substitution is not rescued) is not met, and
 308 we do not make that claim.

309 *Additive substitution.* In both hosts the additive arm cannot be dis-
 310 tinguished from the gateless arm (+0.4, +0.6, +0.1 pp and +0.8, +1.4,
 311 +9.3 pp; all intervals include zero). The point-estimate rescue ratios are
 312 small ($\rho = 0.029$, paired-seed interval $[0.011, 0.059]$; in the fully frozen host
 313 $\rho = 0.194$, $[-0.002, 0.326]$), but because gate removal is only weakened the
 314 pre-specified rule labels the substitution test *not assessable*, and we report it
 315 descriptively.

316 *Paderborn, shared bearings.* The gate-removal cost holds (+7.6, +6.5,
 317 +4.4 pp, intervals excluding zero at all three levels); the partially frozen
 318 block’s advantage over S4D is weakened (+5.6, +6.5, +8.3 pp, resolved at
 319 −6 dB only).

Table 3: Independent training and evaluation noise, Paderborn shared bearings (KA04, KA16, KI04, KI14; 1500 → 900 rpm). Paired differences under final-epoch reporting with 95% t -intervals, seeds positive and pre-specified label. Means: BM3-frozen 81.5 / 83.8 / 85.9, S4D 75.9 / 77.3 / 77.6, frozen−gate 73.8 / 77.3 / 81.5 %.

Contrast	0 dB	−2 dB	−6 dB	label
	+5.6	+6.5	+8.3	
BM3-frozen	[−4.4, 15.5]	[−2.9, 15.9]	[2.7, 13.8]	
− S4D	(4/5)	(4/5)	(5/5)	weakened
	+7.6	+6.5	+4.4	
BM3-frozen	[2.7, 12.6]	[1.7, 11.3]	[3.2, 5.6]	
− frozen−gate	(5/5)	(4/5)	(5/5)	holds

320 4.2. Freezing the input-dependent Δ , A , B , C projections

321 Freezing the input-dependent Δ , A , B , C projections (Section 3.2) costs
 322 at most 3.7 pp anywhere in the adjudication band on the point estimates
 323 under final-epoch reporting (+3.7, +0.9, +0.2 pp at 0, −2, −6 dB), while
 324 the frozen block retains an advantage of +15.5, +19.8, +16.4 pp over S4D
 325 (Table 4, Figure 2). Five seeds bound these quantities loosely: paired 95% in-
 326 tervals for $\Delta(\text{BM3} - \text{frozen})$ are $[-3.8, 11.3]$, $[-4.5, 6.3]$, $[-12.1, 12.6]$ pp (BM3
 327 ahead in 4/5, 3/5, 2/5 seeds), and for $\Delta(\text{frozen} - \text{S4D})$ $[4.3, 26.8]$, $[12.0, 27.7]$,
 328 $[4.1, 28.7]$ pp (5/5, 5/5, 4/5 seeds). The pre-specified rule attributed robust-
 329 ness to selectivity only if $\Delta(\text{BM3} - \text{frozen}) \geq 10$ pp in deep noise with the

Table 4: BM3, BM3-frozen and S4D, macro-F1 (%), mean over 5 seeds; Δ columns are seed-paired mean differences in percentage points. Upper block: final epoch of the 50-epoch budget on the adjudication band (primary; re-runs of Sections 4.3 and 4.10). Middle block: best epoch on the evaluation condition in the same re-runs (sensitivity). Lower block: best epoch in the original frozen-selectivity grid for levels outside the band (auxiliary; not re-run). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3-frozen})$	$\Delta(\text{frozen-S4D})$
<i>Final epoch (primary)</i>					
0 dB	86.9	83.1	67.6	+3.7	+15.5
−2 dB	88.3	87.3	67.5	+0.9	+19.8
−6 dB	83.3	83.1	66.7	+0.2	+16.4
<i>Best epoch (sensitivity, same re-runs)</i>					
0 dB	95.6	90.6	75.9	+5.1	+14.6
−2 dB	95.8	92.3	75.1	+3.5	+17.2
−6 dB	88.6	85.7	68.6	+3.0	+17.1
<i>Auxiliary (best epoch, original grid)</i>					
clean	78.9	78.9	81.0	−0.0	−2.1
−10 dB*	68.1	64.3	52.3	+3.8	+12.0

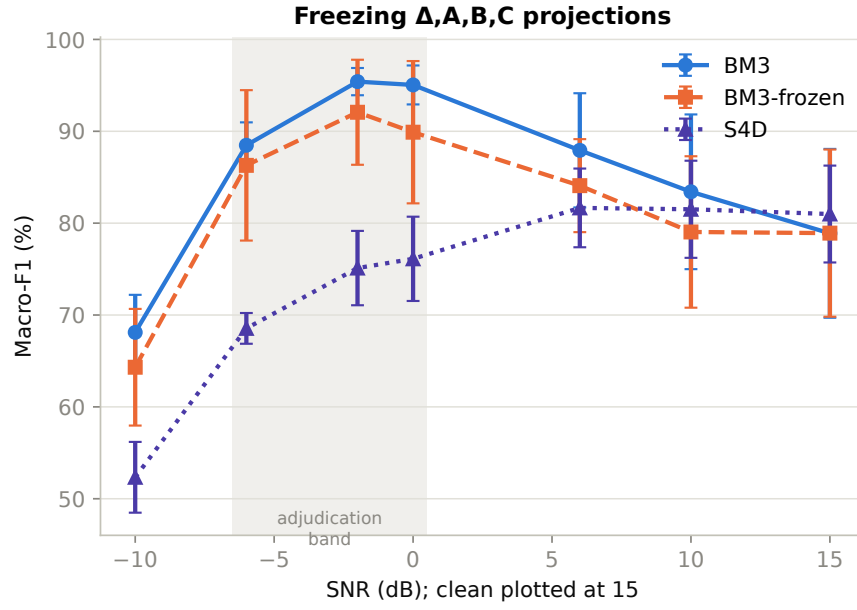


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid (best epoch, original frozen-selectivity grid; final-epoch values on the adjudication band are in Table 4, and levels outside the band are auxiliary). The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band.

330 frozen arm collapsing toward S4D. It was written for best-epoch reporting,
331 under which the original grid gives $\Delta(\text{BM3-frozen}) = +5.1, +3.3, +2.2$ pp
332 (re-run: $+5.1, +3.5, +3.0$) and $\Delta(\text{frozen-S4D}) = +13.8, +17.0, +17.7$ pp.
333 Neither condition holds under either rule, so the pre-specified verdict is that
334 these projections are *not the primary mechanism*. This is a failure to meet
335 a confirmation criterion, not a demonstration that freezing is cheap: the
336 intervals above are compatible with differences larger than 10 pp, and no e-
337 quivalence margin was pre-specified or tested. One limit applies: the frozen
338 arm still contains input-dependent trapezoidal weights and rotation angles, so
339 this result concerns Δ, A, B, C only; Section 4.3 removes that limit. On clean
340 data (auxiliary, best epoch) S4D matches or slightly exceeds both Mamba-
341 style arms—in line with the clean-benchmark comparison of [4], which, like
342 this auxiliary row, involves no added noise. In the noisy regime, the block’s
343 advantage over S4D is not removed by freezing these four terms, so it must
344 be carried by other parts of the block or by the remaining input-dependent
345 scan terms.

346 4.3. Freezing the remaining input-dependent scan terms

347 Table 5 reports the three layered arms of Section 3.2 on the adjudication
348 band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original training
349 code, the same evaluation fingerprint, with the original BM3 arm re-run
350 in the same grid). The question is whether the advantage that survived
351 freezing Δ, A, B, C is carried by the two input-dependent terms the frozen
352 arm retained.

353 It is not. With the scan’s dynamics and read/write coefficients input-
354 independent, *frozen-all* is $+21.9, +24.0$ and $+21.2$ pp above S4D at 0, -2
355 and -6 dB under final-epoch reporting, positive in 5/5 seeds at every level
356 and with intervals excluding zero—if anything a larger margin than the par-
357 tially frozen arm’s $+15.5, +19.8, +16.4$ pp (Table 13). None of the three freez-
358 ing steps costs accuracy: the paired differences BM3-frozen – frozen-trap,
359 – frozen-rot and – frozen-all are negative at all three levels (between -1.8
360 and -6.3 pp under final-epoch reporting), i.e. the more heavily frozen arm is
361 numerically ahead, with every interval including zero. The direction is not
362 an artefact of capacity: *frozen-all* has 103,842 parameters against 112,410
363 for BM3-frozen and 177,938 for BM3. At the deepest level *frozen-all* also
364 exceeds the full BM3 block by 4.5 pp ($[-7.2, -1.9]$ for BM3 – frozen-all, 0/5
365 seeds in BM3’s favour); we report this but do not build on it, since five seeds

Table 5: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen-trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen-rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen-all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen-all − S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen − frozen-all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 − frozen-all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

366 and one speed pair cannot separate a genuine advantage of a time-invariant
367 scan from a variance effect.

368 Two readings of the whole step are therefore available, and the data do not
369 separate them: input-dependence of the scan contributes nothing measurable
370 here, or it contributes something that the learned constants recover under
371 this training budget. Either way the conclusion for the attribution chain
372 is the same and now rests on a complete intervention rather than a partial
373 one: **the Mamba-style block’s deep-noise advantage over a time-**
374 **invariant S4D does not require any input-dependent term in its**
375 **scan.** What remains to carry it is the rest of the block—the convolutional
376 stem, the heavy-tailed A parameterisation, the normalisations and the output
377 gate—which is what Sections 4.4–4.5 examine. The re-run also reproduces
378 the original grids: the BM3 arm’s best-epoch means (95.6, 95.8, 88.6) exceed
379 the original frozen-selectivity grid’s 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp
380 (a different PyTorch/Triton build, Section 4.10).

381 4.4. Removing the gate is the only ablation with a large, one-signed effect

382 “Necessary” is used here in the operational sense of this ablation: remov-
383 ing the gate from the frozen block, with the other components and the budget
384 unchanged, removes most of the block’s advantage. The gate result is large,
385 one-signed and grows with noise depth: under final-epoch reporting removal

Table 6: Single-component removal on the frozen base, $\Delta(\text{frozen} - \text{arm})$ in percentage points: final-epoch mean, 95% t -interval over seeds and number of seeds (of 5) with a positive difference (re-runs of Section 4.10); “best” is the best-epoch value of the original component-ablation grid, for which the thresholds were pre-specified. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
	+9.1 [2.9, 15.3] (5/5); best +13.1	+15.6 [−0.8, 32.0] (4/5); best +21.1	+22.6 [5.8, 39.4] (5/5); best +23.5	PRIMARY (all $\geq$ 8 pp, both rules)
conv stem ($k=7$)	−13.2 [−24.1, −2.2] (0/5); best −6.6	−7.5 [−14.6, −0.3] (0/5); best −2.7	+5.9 [−4.0, 15.8] (3/5); best +7.9	outside bins; regime trade-off
heavy-tailed A → S4D-style A	−1.9 [−9.2, 5.3] (2/5); best −2.4	−2.6 [−8.6, 3.4] (2/5); best −2.2	−4.8 [−16.4, 6.9] (2/5); best −4.2	outside bins; intervals include 0

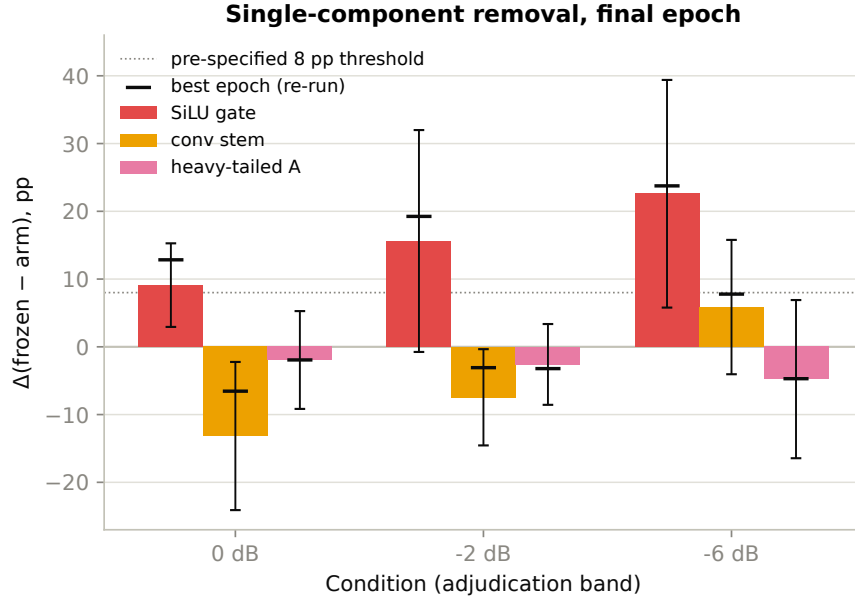


Figure 3: Component-removal Δ (percentage points, frozen $-$ arm) on the adjudication band for the three single-component ablations on the frozen base: final-epoch means with 95% t -intervals over seeds (bars) and best-epoch means of the same re-runs (black ticks). Only gate removal is large and one-signed at every level and clears the pre-specified 8 pp threshold.

costs 9.1, 15.6, 22.6 pp at 0, -2 and -6 dB (paired 95% intervals [2.9, 15.3], $[-0.8, 32.0]$, [5.8, 39.4]; 5/5, 4/5, 5/5 seeds), above the pre-specified 8 pp threshold at every level; the interval at -2 dB includes zero because of the large seed variance of the gateless arm. Under the best-epoch rule for which the threshold was written, the component-ablation grid gives 13.1, 21.1 and 23.5 pp ([6.5, 19.6], [3.8, 38.5], [8.5, 38.6]; 5/5 seeds at every level), so the verdict is *primary* under both rules. The gateless arm gives up most of the Mamba advantage but is not resolved against plain S4D: frozen-gate $-$ S4D is +6.4, +4.2, -6.2 pp, with every interval including zero. We therefore describe it as roughly at the level of S4D, below it only on the point estimate at -6 dB; this suggests, but does not establish, that components interact rather than add. The other two components tell regime stories rather than robustness stories. Removing the convolutional stem is beneficial at 0 and -2 dB (frozen $-$ frozen-conv = -13.2 [$-24.1, -2.2$] and -7.5 [$-14.6, -0.3$], with the stemless arm ahead in every seed) and costly at -6 dB (+5.9, [$-4.0, 15.8$], 3/5 seeds); under best-epoch reporting the pattern was the same ($-6.6, -2.7, +7.9$ pp). On clean cross-speed data (auxiliary, best epoch, original grid) the stem costs 21.0 pp (removal yields 98.8 vs. 77.8, with seed-std collapsing from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal patterns mistransferring across conditions, or with optimisation friction under the matched budget; we report both readings. Replacing the heavy-tailed state parameterisation by an S4D-style one is numerically beneficial at all three levels (by 1.9, 2.6, 4.8 pp under final-epoch and 2.4, 2.2, 4.2 pp under best-epoch reporting), but every paired interval includes zero, so we draw no directional conclusion from it.

4.5. Adding a narrow gate branch to S4D

The pre-specified recovery ratio is $R = \text{mean over the band of (S4D+gate-S4D)/(frozen-S4D)}$. Under final-epoch reporting $R = \mathbf{0.196}$ (per level 0.249, 0.341, -0.003 ; seed-resampled 95% interval [0.09, 0.31]); under the best-epoch rule for which it was specified, $R = 0.174$ ([0.04, 0.29]). The pre-specified bins were $R \geq 0.7$ (confirmed), $0.3 \leq R < 0.7$ (partial recovery) and $R < 0.3$ (insufficient); both values fall in the last, although the final-epoch interval reaches the 0.3 boundary. The per-level paired gains of S4D+gate over S4D under final-epoch reporting are +3.9 ($[-0.8, 8.6]$), +6.8 ([2.4, 11.1]), -0.1 ($[-4.0, 3.9]$) pp. At the deepest adjudicated level the graft recovers nothing. Removal and addition are not mirror-image interventions. In BM3 the gate has its own $64 \rightarrow 128$ projection and multiplies the scan

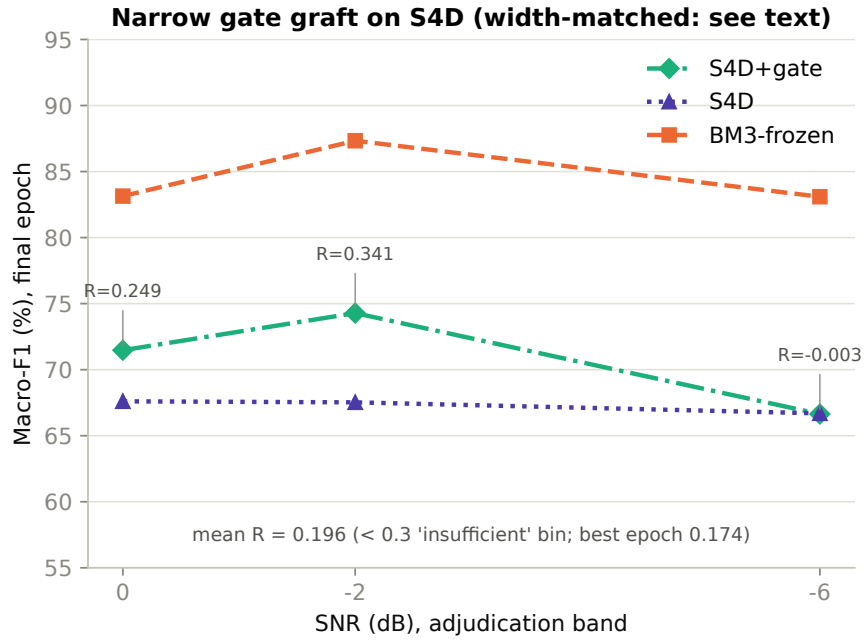


Figure 4: Graft outcome under final-epoch reporting: S4D+gate, plain S4D, and BM3-frozen on the adjudication band (one re-run grid), with the per-level recovery ratio $R = (\text{S4D+gate} - \text{S4D}) / (\text{frozen} - \text{S4D})$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table 8.

Table 7: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). Upper block: final epoch, all three arms from the same re-run (Section 4.10). Lower block: best epoch as pre-specified, with S4D+gate from the graft grid and S4D and BM3-frozen, and hence R , from the component-ablation grid of Section 4.4 (independent runs of the same arms differ by 0.1–0.7 pp). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
<i>Final epoch (primary)</i>				
0 dB	71.5	67.6	83.1	0.249
−2 dB	74.3	67.5	87.3	0.341
−6 dB	66.6	66.7	83.1	−0.003
<i>Best epoch (as pre-specified)</i>				
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	0.035

423 output at the inner width of 128 before the output projection; removing it
 424 deletes 32,768 parameters. In the S4D host the grafted gate is a $64 \rightarrow 64$
 425 linear branch (with bias) computed from the layer input and multiplying the
 426 SSM output before the residual addition; it adds 16,640 parameters over four
 427 layers. The two share the multiplicative SiLU form but not width, projec-
 428 tion structure or parameter count. Removal costs 9–23 pp in the frozen block,
 429 whereas this graft changes S4D by between −0.1 and +6.8 pp (final epoch).
 430 What this shows is that *this gate branch, in this host and under this budget,*
 431 *recovers little*. Three readings fit the data and cannot be separated here: (i)
 432 host dependence, i.e. the gate needs the stem or scan context it co-trained
 433 with; (ii) implementation differences in width, projection and placement; (iii)
 434 transplant friction, i.e. the grafted branch is harder to optimise in a foreign
 435 host under the matched budget (training converges normally, but convergence
 436 is not robustness). Section 4.6 tests reading (ii) with a width-matched graft
 437 and an equal-capacity non-multiplicative control; a tuning or longer-budget
 438 comparison selected on a source-domain validation set was not run.

439 The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) -$
 440 $F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under
 441 last-epoch reporting $I = 5.2$ [2.3, 8.1], 8.9 [−6.2, 23.9] and 22.6 [2.9, 42.4] pp

at 0, -2 and -6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 [1.4, 14.2], 14.4 $[-1.4, 30.3]$ and 23.1 [8.7, 37.4] pp. A positive I means the gate adds more on the frozen host than on S4D. Because the host differs in several respects at once and the two gates are not aligned, I cannot be assigned to a specific component or interaction term, and the seed pairing across arms is nominal (arms share seed indices, not initialisations).

4.6. A width-matched graft and an additive control

The graft of Section 4.5 is narrower and smaller than the donor gate. Two further arms rebuild it in the donor’s shape (Table 1). In each layer the layer input is projected to x and z of width 128 (the frozen block’s inner width), the S4D layer runs on x at width 128 with state size 64—so that its own parameter count and total state size equal the host’s 64-wide, state-128 layer—and an output projection maps back to width 64. *S4D+gate* (*wm*) combines $y \odot \text{SiLU}(z)$, as the donor does; *S4D+branch* uses the same modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938), which biases the test towards recovery. A construction check with forward hooks confirms that the tensor entering the output projection equals the intended formula for each arm and not the other, and that the two arms are identical at initialisation. The thresholds were committed publicly before the grid ran (Section 3.5): R as before, under final-epoch reporting; the verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a bin boundary, in which case we write at the tier less favourable to the claim; and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an interval excluding zero.

The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159, 0.298, 0.431). Its interval, $[0.16, 0.52]$, crosses the 0.3 boundary, so the pre-specified verdict is INCONCLUSIVE between “insufficient” and “partial”, and we report it at the less favourable tier: a graft built in the donor’s shape recovers part of the gap. The recovery is not gate-specific. The additive branch, with the same parameters, recovers at least as much ($R = 0.408$), $\Delta R = -0.112 [-0.26, 0.02]$, and at 0 dB the gated arm is below the additive one by 3.0 pp $[-4.5, -1.5]$, 0/5 seeds). With frozen—all in the denominator the gated arm’s ratio is 0.231 [0.13, 0.33]; under best-epoch reporting the two ratios are 0.459 and 0.483. What a graft recovers on this transition therefore follows the added structure—a wider channel axis and input/output

Table 8: Width-matched graft and additive control on the original transition. Macro-F1 (%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of (arm – S4D)/(BM3-frozen – S4D), with a seed-resampled 95% interval. S4D, BM3-frozen and the narrow graft are the re-runs of Section 4.10. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB	–2 dB	–6 dB	R	95% interval
S4D	67.6	67.5	66.7	–	–
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (wm)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	–
$\Delta R = \text{S4D+gate (wm)} - \text{S4D+branch}$				–0.112	[–0.26, 0.02]

479 projections—rather than the multiplicative interaction. Reading (ii) of Sec-
480 tion 4.5 is partly confirmed, in that the narrow graft understated what added
481 structure can recover; but in the S4D host the gate itself contributes noth-
482 ing beyond an equal-capacity additive branch, whereas removing it from the
483 frozen Mamba-3 block costs 9–23 pp under the same rule (Table 13).

484 4.7. Replacing the gate in its native host

485 The previous subsection changes the host and keeps the operator. A pre-
486 specified grid (15 cells, original transition) does the converse: it keeps the host
487 and changes the operator. In *frozen+add* the scan kernel of every BM3-frozen
488 layer is called without gating and the layer output becomes $W_{\text{out}}(y + \text{SiLU}(z))$;
489 modules, parameter count (112,410) and initial weights are bitwise those of
490 BM3-frozen, and a construction check confirms that the donor operator is
491 $y \odot \text{SiLU}(z)$ and that the new arm computes $y + \text{SiLU}(z)$. The pre-specified
492 statistic is the native rescue ratio $\rho = (\text{add} - \text{no gate})/(\text{frozen} - \text{no gate})$,
493 mean over levels, final epoch, with a seed-resampled interval; *not rescued*
494 requires $\rho < 0.3$ with an upper bound below 0.5.

495 The additive branch does not measurably rescue the loss. *frozen+add*
496 cannot be distinguished from the gateless arm at any level (+0.6, –0.2,
497 +0.4 pp, every interval including zero), and the multiplicative arm exceeds
498 it by 8.5, 15.8, 22.2 pp. The rescue ratio is $\rho = 0.024$ (per level 0.070,
499 –0.015, 0.016), with a 95% interval of [–1.159, 0.486] under the pre-specified
500 bootstrap, which resampled each arm and level independently; resampling
501 seeds jointly across arms and levels, as the paired design warrants, gives

Table 9: Gate replaced by an additive branch inside BM3-frozen, original and reverse transitions. Macro-F1 (%), mean over 5 seeds; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

	0 dB		−2 dB		−6 dB	
Arm	best	final	best	final	best	final
<i>Original transition (37.5 Hz/11 kN → 40 Hz/10 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	90.6	83.1	92.3	87.3	85.7	83.1
frozen+add ($y + \text{SiLU}(z)$)	78.0	74.7	72.6	71.5	62.3	60.9
frozen−gate (no z)	77.7	74.0	73.1	71.7	61.9	60.5
	+8.5		+15.8		+22.2	
BM3-frozen − frozen+add	[1.1, 15.9] (5/5)		[−1.2, 32.9] (4/5)		[4.6, 39.9] (5/5)	
	+0.6		−0.2		+0.4	
frozen+add − frozen−gate	[−1.7, 3.0] (3/5)		[−3.4, 3.0] (3/5)		[−0.7, 1.4] (3/5)	
<i>Reverse transition (40 Hz/10 kN → 37.5 Hz/11 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	81.1	74.7	84.0	76.7	80.8	73.9
frozen+add ($y + \text{SiLU}(z)$)	64.3	56.5	62.2	54.4	55.7	50.5
frozen−gate (no z)	63.8	56.2	62.0	54.7	57.1	50.0
	+18.1		+22.3		+23.4	
BM3-frozen − frozen+add	[1.2, 35.0] (5/5)		[3.5, 41.0] (5/5)		[6.8, 40.0] (5/5)	
	+0.3		−0.3		+0.6	
frozen+add − frozen−gate	[−0.6, 1.2] (3/5)		[−1.0, 0.4] (2/5)		[−0.2, 1.4] (5/5)	

502 $[-0.043, 0.136]$. Under best-epoch reporting $\rho = 0.005$. By the pre-specified
503 rule the verdict is *not rescued*: the point-estimate rescue is small and below
504 the pre-specified criterion, which is not a demonstration that the two arms
505 are equivalent.

506 A further pre-specified grid (15 cells) repeats the substitution on the
507 reverse transition (Section 4.8), with BM3-frozen and frozen+gate taken from
508 the reverse grid. The outcome is the same (Table 9, lower block): frozen+add
509 cannot be distinguished from the gateless arm at any level (+0.3, -0.3,
510 +0.6 pp, every interval including zero), and the multiplicative arm exceeds it
511 by +18.1, +22.3, +23.4 pp (intervals excluding zero, 5/5 seeds at every level).
512 The rescue ratio is $\rho = 0.009$ (per level 0.017, -0.014, 0.023), with a 95%
513 interval of $[-0.943, 0.362]$ under the pre-specified bootstrap and $[0.002, 0.018]$
514 under paired-seed resampling; under best-epoch reporting $\rho = -0.005$. The
515 pre-specified verdict is again *not rescued*.

516 Together with Section 4.6 this gives the two halves of a symmetric test.
517 In the S4D host the gate did not recover more than an additive branch of
518 the same size on the original transition and recovered somewhat more on the
519 reverse one; in the Mamba-3 host an additive branch of the same size could
520 not be distinguished from removing the gate, on both transitions. Under the
521 earlier noise protocol this pattern is specific to the multiplicative operator
522 in its native host; Section 4.1 shows that the size of the gate’s contribution
523 there depends on noise depth.

524 4.8. The reverse transition

525 A pre-specified grid repeats the chain’s central contrasts on the reverse
526 transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN (75 cells: frozen+all, BM3-frozen,
527 frozen+gate, S4D and S4D+gate (wm), at three levels and five seeds; Sec-
528 tion 3.1 for the data). This direction is harder for the time-invariant model:
529 on its evaluation set, predicting the majority class gives a macro-F1 of 46.8%
530 and uniform random guessing 41.5%, and S4D sits close to those levels (Ta-
531 ble 10).

532 The first two steps replicate. The fully input-independent scan exceeds
533 S4D by +26.3, +25.6 and +22.2 pp at 0, -2 and -6 dB, with intervals ex-
534 cluding zero and 5/5 seeds at every level, so by the pre-specified rule (at
535 least two of three levels) the first step is replicated on the reverse transition.
536 Removing the gate from BM3-frozen costs 18.4, 22.0 and 24.0 pp, above the
537 8 pp threshold at all three levels, with intervals excluding zero and 5/5 seeds,
538 and the gateless arm is indistinguishable from S4D (intervals include zero).

Table 10: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch and final-epoch reporting; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm (best / final)	0 dB	-2 dB	-6 dB
frozen-all	86.3 / 77.5	87.8 / 77.9	81.7 / 74.0
BM3-frozen	81.1 / 74.7	84.0 / 76.7	80.8 / 73.9
S4D+gate (wm)	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
frozen-gate	63.8 / 56.2	62.0 / 54.7	57.1 / 50.0
S4D	61.2 / 51.2	61.0 / 52.4	54.6 / 51.8
	+26.3	+25.6	+22.2
frozen-all - S4D	[21.5, 31.0] (5/5)	[21.1, 30.1] (5/5)	[18.8, 25.6] (5/5)
	+18.4	+22.0	+24.0
BM3-frozen - frozen-gate	[1.2, 35.6] (5/5)	[3.4, 40.6] (5/5)	[7.6, 40.3] (5/5)
	+5.0	+2.4	-1.8
frozen-gate - S4D	[-17.6, 27.7] (2/5)	[-19.2, 23.9] (2/5)	[-18.9, 15.3] (1/5)
	+21.7	+21.5	+20.5
S4D+gate (wm) - S4D	[15.4, 28.0] (5/5)	[17.0, 26.0] (5/5)	[15.7, 25.2] (5/5)

539 On the third step, which the pre-specification treats as descriptive here, the
 540 width-matched graft recovers most of the gap, $R = 0.911$ [0.82, 1.04] (0.863
 541 with frozen-all in the denominator). A further pre-specified grid (15 cells)
 542 adds the same-size additive control in this direction: it reaches 67.8, 72.1,
 543 68.4% (final epoch) and recovers $R = 0.758$ [0.66, 0.88]. The gate arm ex-
 544 ceeds it by +5.1, +1.7, +3.8 pp (only the -6 dB interval excludes zero),
 545 and the difference in recovery is $\Delta R = 0.153$ [0.06, 0.26]; under best-epoch
 546 reporting it shrinks to 0.037 [-0.1, 0.17]. By the pre-specified descriptive
 547 labels this is *unresolved*: in this direction the gate carries a small increment
 548 over the additive branch, whereas on the original transition it carried none
 549 ($\Delta R = -0.112$). Most of the reverse-direction recovery is therefore available
 550 to an additive branch of the same size, and the remaining gate increment is
 551 direction-dependent. Both ratios rest on an S4D baseline close to the trivial
 552 floor (51–52% against 46.8% for the majority class). Both statements carry
 553 the limits of this pair: the same eight bearings with their roles swapped, and
 554 a single inner-race bearing in the evaluation set.

555 4.9. External validity: two settings on a second dataset

556 A pre-specified grid (75 cells; thresholds, a ceiling-and-floor check and the
 557 writing consequences committed before the first cell) repeats the three central
 558 contrasts on the Paderborn KAt data [16], using real-damage bearings only.
 559 Models train on two outer-race (KA04, KA16) and two inner-race bearings
 560 (KI04, KI14) at 1500 rpm and are evaluated on five different bearings (KA15,
 561 KA22; KI17, KI18, KI21) at 900 rpm, with load and radial force unchanged;
 562 the vibration channel is resampled to 25.6 kHz so that a window spans 80 ms
 563 as in XJTU-SY, and everything else (noise, training, arms, budget) is as in
 564 the main grids. The pre-specified smoke check (two arms, one extra seed)
 565 passed its three criteria, the learnability criterion by 0.3 pp. A second pre-
 566 specified grid (45 cells, three arms) then used the same four bearings for
 567 training and evaluation, at the same two speeds, from different recordings.
 568 This changes the evaluation bearings, their damage profiles, the number of
 569 windows and the class balance together, so the two settings are two protocols
 570 rather than a single-factor manipulation.

571 None of the pre-specified contrasts replicates (Table 11). The fully input-
 572 independent block does not exceed S4D (frozen—all — S4D: -1.4 , -2.5 ,
 573 -3.3 pp, the interval at -6 dB excluding zero in the negative direction), nor
 574 does BM3-frozen (-3.8 , -3.6 , -3.4 pp). Removing the gate does not cost ac-
 575 curacy: BM3-frozen — frozen—gate is -5.0 , -4.6 , -3.3 pp, below zero in ev-
 576 ery seed at every level. With no gate effect to rescue, the native-substitution
 577 test is not assessable; the additive and gateless arms again cannot be dis-
 578 tinguished (0.0 , 0.0 , -0.1 pp). The pre-specified labels are therefore *not*
 579 *replicated* for both freeze contrasts and for gate removal, and *not assessable*
 580 for the substitution test.

581 Two observations bound how this result can be read, and neither changes
 582 the labels. First, the testbed discriminates poorly: every arm ends between
 583 47.7 and 56.9% macro-F1 at every level (majority-class predictor 37.5%, a
 584 balanced random predictor about 50%), all arms fit the training condition
 585 (the gated arms reach the lowest training loss), and none transfers far above
 586 the level reached after the first epoch. Second, where differences are resolved
 587 they run against the Mamba-3 arms, which reach lower training loss and
 588 transfer slightly worse. In the disjoint setting, therefore, the block’s advan-
 589 tage under matched noise and drift is not observed; we do not know whether
 590 this reflects a property of the PU transition (speed reduction to 900 rpm with
 591 disjoint bearings), the protocol transferred unchanged from XJTU-SY, or a

Table 11: Paderborn, real-damage bearings, 1500 rpm $\rightarrow$ 900 rpm. *Disjoint*: training on KA04, KA16, KI04, KI14 and evaluation on KA15, KA22, KI17, KI18, KI21. *Shared*: training and evaluation on KA04, KA16, KI04, KI14 (different recordings). Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds (majority-class macro-F1: 37.5% disjoint, 33.3% shared). Lower block: paired differences under final-epoch reporting, 95% t -interval and seeds (of 5) with a positive difference. “-”: arm not run in that setting. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm / contrast	Disjoint bearings			Shared bearings		
	0 dB	-2 dB	-6 dB	0 dB	-2 dB	-6 dB
BM3-frozen	51.9	50.3	47.7	84.5	85.8	86.6
frozen-all	54.2	51.3	47.8	-	-	-
S4D	55.6	53.9	51.1	76.1	77.3	77.9
frozen-gate	56.9	54.9	51.1	75.5	78.1	81.3
frozen+add	56.9	54.8	51.0	-	-	-
	-3.8	-3.6	-3.4	+8.4	+8.5	+8.7
BM3-frozen	[-11.9, 4.3]	[-12.6, 5.4]	[-8.9, 2.1]	[-1.3, 18.1]	[0.2, 16.8]	[4.3, 13.0]
- S4D	(1/5)	(2/5)	(1/5)	(4/5)	(5/5)	(5/5)
	-5.0	-4.6	-3.3	+9.0	+7.7	+5.3
BM3-frozen	[-8.8, -1.3]	[-9.2, 0.1]	[-7.1, 0.4]	[5.2, 12.8]	[5.0, 10.4]	[2.4, 8.2]
- frozen-gate	(0/5)	(0/5)	(0/5)	(5/5)	(5/5)	(5/5)
	-1.4	-2.5	-3.3			
frozen-all	[-4.0, 1.2]	[-5.3, 0.2]	[-5.8, -0.8]			
- S4D	(1/5)	(1/5)	(0/5)	-	-	-
	0.0	0.0	-0.1			
frozen+add	[-0.9, 0.8]	[-0.4, 0.3]	[-0.4, 0.2]			
- frozen-gate	(2/5)	(3/5)	(1/5)	-	-	-

limit of the XJTU-SY result itself, which the shared-bearing setting below separates.

Shared bearings. With bearing identity shared between training and evaluation, the same condition pair becomes transferable: every arm ends between 75.5 and 86.6% at every level, and the pre-specified transferability gate (5-seed mean $\geq 65\%$ with one-sided 95% lower bound $> 60\%$ at two of three levels for at least one arm) is passed by all three arms at all three levels (smallest lower bound 71.3%). Within this setting both effects replicate under the pre-specified rule (Table 11, right). BM3-frozen exceeds S4D by +8.4, +8.5, +8.7 pp at 0, -2, -6 dB, with intervals excluding zero at -2 and -6 dB; removing the gate costs +9.0, +7.7, +5.3 pp, with intervals excluding zero at all three levels and in every seed; no level runs in the opposite direction. The gateless arm returns to the level of S4D (frozen-gate - S4D: -0.6, +0.8, +3.4 pp, all intervals including zero), as on XJTU-SY. Under best-epoch reporting the freeze contrast is resolved at all three levels (+6.9, +10.8, +13.7 pp) and gate removal only at -6 dB (+1.5, +3.7, +6.4 pp). The cost of gate removal reaches the 8 pp threshold used on XJTU-SY at one level only; we compare directions, not magnitudes, across datasets.

Read together, the two protocols give different results: with shared bearings every arm, S4D included, ends 18.6–38.8 pp higher than with disjoint bearings (independent seeds; per-arm Welch intervals all exclude zero), and the block’s advantage and the cost of gate removal reappear. Because the evaluation bearings, class balance and sample size change together, this does not isolate bearing overlap as the cause. With two bearings per class in the shared setting, a bearing’s identity also determines its label, so a model can succeed partly by recognising bearing-specific signatures. XJTU-SY differs from both Paderborn settings: its training and evaluation bearings are disjoint in each direction, and the block transfers there. The shared-setting contrasts were re-run under independent noise (Section 4.1); *frozen-all* was not run on Paderborn.

4.10. Sensitivity to the reporting rule

The main tables use the final epoch of the 50-epoch budget (Section 3.3). The original grids recorded instead the best epoch on the evaluation condition, and their pre-specified thresholds were written for that rule. To measure how much the rule shapes the conclusions, every arm of the chain was re-run over the adjudication band with per-epoch evaluation, using the original training code and the same evaluation fingerprint: the four arms

629 BM3-frozen, S4D, frozen-gate and S4D+gate (60 cells), the BM3 arm a-
630 longside the layered freezes (Section 4.3), and the capacity control and the
631 two other single-component removals, S4D-wide, frozen-conv and frozen-A
632 (45 cells, pre-specified with a rule for changes of sign). The re-runs used a dif-
633 ferent PyTorch/Triton build from the original grids; their best-epoch values
634 reproduce the original arm-by-condition means to within -0.6 to $+1.4$ pp
635 across the eight re-run arms (the largest deviation is in the high-variance
636 frozen-gate arm at -2 dB).

Table 12: Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting in the re-runs, for every arm of the chain on the adjudication band. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		-2 dB		-6 dB	
	best	final	best	final	best	final
BM3	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
S4D-wide	77.8	67.3	73.3	66.5	69.0	67.1
frozen-gate	77.7	74.0	73.1	71.7	61.9	60.5
frozen-conv	97.1	96.3	95.4	94.8	77.9	77.2
frozen-A	92.5	85.1	95.6	89.9	90.4	87.9
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

637 Best-epoch values exceed final-epoch values by 0.6 – 10.6 pp across the 24
638 arm-by-level means, so best-epoch absolute levels are optimistic. The re-
639 duction is of similar size for the arms being contrasted (e.g., BM3-frozen
640 -7.4 , -5.0 , -2.6 pp and S4D -8.3 , -7.6 , -1.9 pp at 0 , -2 , -6 dB), and the
641 contrasts that carry the argument are correspondingly stable (Table 13).

642 *Selectivity.* The frozen block’s advantage over S4D is $+15.5$, $+19.8$ and
643 $+16.4$ pp (and $+21.9$, $+24.0$, $+21.2$ pp for the fully frozen scan, Section 4.3),
644 versus $+14.6$, $+17.2$ and $+17.1$ pp under best-epoch reporting in the re-run.
645 It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero
646 at every level. *Gate.* Removing the gate costs 9.1 , 15.6 and 22.6 pp (12.8 to
647 23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified
648 threshold at every level; the interval at -2 dB includes zero ($[-0.8, 32.0]$ pp,
649 4/5 seeds) because of the large seed variance of the gateless arm (standard de-

Table 13: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Contrast	0 dB	−2 dB	−6 dB
	+15.5	+19.8	+16.4
BM3-frozen − S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen − frozen−gate	[2.9, 15.3] (5/5)	[−0.8, 32.0] (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	−6.2
frozen−gate − S4D	[−2.9, 15.8] (4/5)	[−14.5, 22.9] (3/5)	[−23.5, 11.2] (1/5)
	+3.9	+6.8	−0.1
S4D+gate − S4D	[−0.8, 8.6] (4/5)	[2.4, 11.1] (5/5)	[−4.0, 3.9] (2/5)

650 viation 14.6 pp at −2 dB). *Gateless versus S4D*. Under final-epoch reporting
651 the gateless arm is not below S4D at 0 and −2 dB (point estimates +6.4 and
652 +4.2 pp) and is below it at −6 dB (−6.2 pp); all three intervals include zero.
653 The statement that gate removal “falls below S4D” is therefore supported
654 only as a point-estimate observation at the deepest level, and we phrase it as
655 “to or below” elsewhere. *Graft*. The recovery ratio is 0.249, 0.341 and −0.003
656 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the
657 pre-specified “insufficient” bin on the point estimate. The seed-resampled
658 95% interval is [0.09, 0.31] and so reaches the 0.3 boundary: with five seeds
659 the data support “far below confirmation (0.7) and probably below 0.3”, not
660 a precise value of R . The graft gain over S4D is resolved at −2 dB only
661 (+6.8 pp, [2.4, 11.1]) and is indistinguishable from zero at −6 dB.

662 *Remaining arms*. By the rule pre-specified for the 45-cell grid, no arm
663 changes sign between the two rules at any level where both values exceed
664 1 pp, so all main-table entries are reported under the final-epoch rule with
665 unchanged wording. The capacity control holds: BM3-frozen exceeds S4D-
666 wide by +15.9, +20.8, +16.0 pp (intervals excluding zero at all three levels),
667 and S4D-wide − S4D is −0.4, −1.0, +0.4 pp (intervals including zero). For
668 the stem the final-epoch rule sharpens the regime trade-off (intervals at 0
669 and −2 dB now exclude zero) without reversing it.

670 Because the final epoch of the same evaluation condition has been in-
671 spected repeatedly during this study, last-epoch values are also not a fresh
672 confirmatory test. The reporting rule therefore changes absolute levels and
673 the strength of the gateless-versus-S4D statement, but not the ordering BM3-

frozen > {S4D+gate, frozen+gate, S4D}, the size of the frozen block’s advantage over S4D, or the verdicts of the pre-specified gate and graft rules.

4.11. The interaction effect concentrates on the hard class

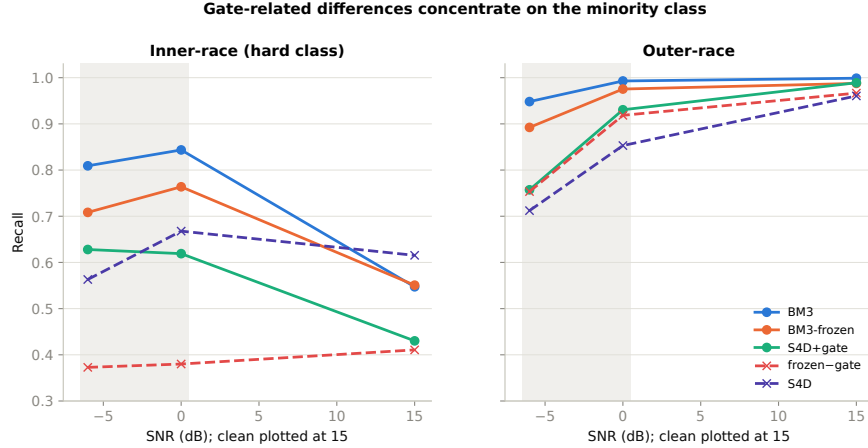


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch selection on the evaluation condition: 5 arms $\times$ {clean, 0, -6 dB} $\times$ 5 seeds, per-sample predictions dumped and re-verified against reported metrics cell-by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate presence: in all three gated arms IR recall at -6 dB exceeds its clean value (e.g., 0.55 $\rightarrow$ 0.81 for BM3), while in both gateless arms it is below its clean value (frozen+gate: 0.41 $\rightarrow$ 0.37; S4D: 0.62 $\rightarrow$ 0.56, non-monotone, rising to 0.67 at 0 dB) (Figure 5). Both ablation directions hit IR hardest—gate removal costs IR 2.3–3.8 $\times$ more F1 than OR at every level (F1 from predictions pooled over seeds), and the graft’s noise-time gains also concentrate on IR (-6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages the two classes, the arm-level robustness gaps of Sections 4.2–4.5 are carried substantially by the hard class. The gate-related differences are therefore class-concentrated rather than uniform, and fall on the class with the lowest recall. We did not measure classification margins, so we do not describe IR as a “low-margin” class. Because this grid is separate from the main 50-epoch

694 grid and the IR class comes from a single training bearing (Section 3.1), these
 695 results are descriptive and do not decompose the main-grid effects.

696 5. Discussion

697 **What can be credited?** The ablation chain supports a narrow, con-
 698 ditional statement: within this Mamba-3 block, on this testbed and with
 699 noise-aware training, (i) making the scan fully input-independent—first the
 700 Δ , A , B , C projections, then the trapezoidal weight and rotation angles as
 701 well—did not remove the block’s deep-noise advantage over a time-invariant
 702 S4D, and at the deepest level the fully frozen arm matched or exceeded the
 703 original block with fewer parameters than BM3 or BM3-frozen (though more
 704 than S4D), (ii) removing the output gate removed most of that advantage—
 705 (i) and (ii) holding on the reverse transition as well—and (iii) adding the
 706 gate to S4D recovered part of the advantage only together with added width
 707 and projections, and an additive branch of the same size recovered as much,
 708 while (iv) in the Mamba-3 block an additive branch of the same size could not
 709 be distinguished from removing the gate, on both transitions. Under inde-
 710 pendent training and evaluation noise (Section 4.1), (i) holds, (ii) is resolved
 711 only at -6 dB, and in the fully frozen block the gate carries the advantage at
 712 -6 dB but not at 0 and -2 dB. These observations are consistent with a gate
 713 benefit that is specific to the multiplicative operator in its native host and
 714 grows with noise depth. They do not identify which other component the
 715 gate depends on. Nor do they separate “input-dependence contributes noth-
 716 ing here” from “what it contributes is recovered by learned constants under
 717 this budget”; the two readings are indistinguishable in our data and lead to
 718 the same attribution. Representability results for selectivity [8] are not con-
 719 tradicted by a single ablation on one testbed. The S4D-superiority finding [4]
 720 was obtained on clean classification benchmarks, without matched noise or a
 721 shift of operating condition, so it does not speak to the regime studied here
 722 and we do not read it as covering it. Two points of contact are nevertheless
 723 consistent with it: our clean-data parity between S4D and the Mamba-style
 724 arms, and our finding that input-dependent selectivity is not needed for the
 725 block’s advantage. What this paper adds is specific to the matched-noise,
 726 cross-condition regime: there S4D trails the frozen Mamba-3 block by 15–
 727 26 pp under final-epoch reporting (partial and full freeze, both transitions),
 728 and the gap does not require input-dependent scan coefficients; at the deep-
 729 est level it depends on the multiplicative output gate. This is a statement

730 about a different regime, not a contradiction of the clean-benchmark result.

731 **Implications for backbone selection and for transplant practice.**

732 For practitioners choosing backbones for noisy, drifting signal regimes, the
733 results suggest that the block’s benefit should be judged as a composition,
734 and that a gate added to a different host should not be expected to reproduce
735 it: in our S4D host a gate built in the donor’s shape recovered no more than
736 an additive branch of the same size on the original transition (Section 4.6)
737 and only slightly more on the reverse one (Section 4.8), and neither design
738 was tuned. A hypothesis for future work follows: a recomposed block guided
739 by the component map (no conv stem, gate retained, S4D-style A) may match
740 or exceed the original on this testbed; we leave it untested, because post-hoc
741 recomposition after seeing the map would not carry the evidential status of
742 the pre-specified chain.

743 **Why might gating be composition-bound?** We refrain from a mech-
744 anistic claim beyond the data, but note three compatible observations. First,
745 the gate-related differences concentrate on the minority, lowest-recall class
746 (Section 4.11), and matched-noise training lifts that class *only when the gate*
747 *is present*. One hypothesis, which we did not test, is that the gate is the
748 component through which matched-noise training improves hard-class recall,
749 i.e., that it exploits noise-aware training rather than acting as a test-time de-
750 noiser; we measured neither classification margins nor gate activations, and
751 the reading is in any case bounded by the matched-SNR scope (Section 6).
752 Second, the multiplicative form admits a per-sample suppression interpreta-
753 tion, but no gate statistics were recorded, so this remains a hypothesis
754 rather than an observation. Third, the gateless composition falling to or
755 below S4D at the deepest noise levels is compatible with the other compo-
756 nents being calibrated to a gated regime (e.g., the stem may pass features
757 that are only safe when the gate can veto them); this is the weakest of the
758 three observations, since the gateless–S4D gap is not resolved with five seeds
759 (Section 4.10). All are testable in future work via pairwise interaction grids
760 and mismatched-noise protocols.

761 6. Boundaries and Limitations

762 We state the scope plainly.

- 763 1. **Testbed scope.** All claims are bounded to bearing vibration from
764 one pair of operating conditions of one dataset (37.5 Hz/11 kN and

40 Hz/10 kN; speed and load both change), binary classification, GPU execution (no edge/real-time claims). Within each direction the training and evaluation bearings are disjoint; the pair is run in both directions, but the reverse direction uses the same eight bearings with their roles swapped and has a single IR bearing in its evaluation set (the forward direction a single IR bearing in its training set), so it tests direction sensitivity rather than replicating on new bearings; the third condition of the dataset has no IR bearing. The five seeds do not resample bearings (Section 3.1), so the intervals carry no bearing-level uncertainty and class-level results may reflect individual bearings. Further independent datasets and multi-class settings are future work.

2. **Attribution level.** Credit assigned to “the rest of the composition” is by elimination within the enumerated component set; pairwise interaction terms are not decomposed, and the host-by-gate contrast of Section 4.5 is a four-arm estimate that cannot be assigned to a specific component.
3. **The transplant conclusion is budget-conditional.** All arms share a matched 50-epoch budget; the grafted gate trains normally yet recovers little, and convergence is not robustness (Section 4.5)—but we cannot exclude that a longer or graft-tuned budget would recover more. Our statement is therefore limited to *this gate branch, in this host, under a matched training budget*.
4. **Matched-SNR scope.** Training and evaluation share the same injected SNR throughout (Section 3.1); “robustness” in this paper is matched-noise robustness under noise-aware training, not clean-train/noisy-test generalisation. The gate’s benefit may operate by exploiting matched-noise training (consistent with Section 4.11) rather than by test-time denoising; attribution under mismatched-noise protocols is untested and may differ.
5. **Graft dual reading.** Interaction dependence vs. transplant friction (Section 4.5), the latter now subsumed under the budget condition of item 3.
6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a TCN baseline slightly overtakes the Mamba-style block, and -10 dB is near-chance for the time-invariant baseline and excluded from adjudication; the demonstrated advantage regime is matched, Gaussian-dominated noise.

- 802 7. **Conv-stem dual reading.** Its clean-data cost admits both representation-
803 mistransfer and optimisation-difficulty readings, reported in place (Sec-
804 tion 4.4).
- 805 8. **Reporting rule.** Reported values are final-epoch macro-F1 under a
806 fixed 50-epoch budget. The original grids’ best-epoch rule, for which
807 the thresholds of the early grids were pre-specified, is reported as a
808 sensitivity analysis and is optimistic by 0.6–10.6 pp (Section 4.10). Ev-
809 ery arm of the chain was re-run on the three adjudication levels; the
810 +10/+6/clean/−10 dB levels and the per-class grid were not re-run
811 and are auxiliary (best epoch). No source-domain validation rule was
812 available, and because the final epoch of the same evaluation condi-
813 tion has been inspected repeatedly, final-epoch values are not a fresh
814 confirmatory test.
- 815 9. **Scope of the freezing intervention.** The chain’s first step freezes
816 the input-dependent parts of Δ , A , B , C only; the fully input-independent
817 scan is a separate grid (Section 4.3) on the three adjudication levels,
818 which was specified after the earlier grids had been seen. No equivalence
819 margin was pre-specified for the cost of freezing, so “no measurable
820 cost” means the point estimates favour the frozen arms with intervals
821 including zero, not a demonstrated equivalence. Freezing is implement-
822 ed inside this Mamba-3 layer; a conclusion about input-dependence in
823 general would need other blocks and tasks.
- 824 10. **Gate interventions.** Under independent noise, gate removal is re-
825 solved only at −6 dB in both the partially and the fully frozen host,
826 and without its gate the fully frozen block still exceeds S4D at 0 and
827 −2 dB, so the combined statement that the fully frozen block’s advan-
828 tage is carried by the gate is not supported. The additive substitu-
829 tion could not be distinguished from gate removal in either host; with
830 gate removal only weakened, the pre-specified rule leaves it not assess-
831 able under independent noise (*not rescued* under the earlier protocol,
832 in both directions). In S4D the width-matched arms are larger than
833 the donor block, and the gate’s increment over the additive branch is
834 direction-dependent (paired-seed interval for the reverse-direction dif-
835 ference [0.013, 0.338]).
- 836 11. **Noise protocol.** All grids before Section 4.1 used training and evalua-
837 tion noise shared by index; their arm means differ from the independent-
838 noise re-run by at most 4 pp where both exist, but grids that were not
839 re-run (component ablations of the stem and A , grafts, reverse transi-

- tion, per-class grid) carry that protocol.
12. **Strength of the pre-specification evidence.** Thresholds were pre-specified internally; for all but the last four follow-up grids the supporting evidence is local file times and run-directory names, not an external time-stamped registry (Section 3.5). The process-level repeat check is a single observation.
 13. **Construction checks.** For the frozen arm the state-dictionary check compares names and shapes of non-target tensors; value-level comparison was applied to the component-ablation and graft arms only.
 14. **Second dataset.** On Paderborn, the gate-removal cost replicates when training and evaluation share bearings (also under independent noise), the partially frozen block’s advantage only at -6 dB, and nothing replicates on bearings disjoint from the training set, where every examined architecture, S4D included, transferred poorly (47.7–56.9% against a balanced-chance level of about 50%). The two Paderborn protocols differ in more than bearing overlap, so they do not identify why transfer fails; *frozen-all* was not tested on Paderborn.
 15. **Sample size.** Every cell mean uses five seeds. The smallest attainable one-sided Wilcoxon p is 0.031, paired intervals are wide, and several secondary statements (conv stem, A parameterisation, gateless versus S4D) are not resolved at this size; conclusions rest on effect sizes that clear the pre-specified thresholds, not on significance tests, which were never adjudication criteria.

A single-repeat check of process-level non-determinism (~ 0.7 pp) and the label-hash/adjudication trail are documented in the reproducibility appendices (Appendix A–Appendix B).

7. Conclusion

In this Mamba-3 block, on one controlled shift $\times$ noise testbed with matched-noise training, making the scan’s dynamics and read/write coefficients input-independent did not remove the block’s advantage over a time-invariant S4D; under independent training and evaluation noise the fully frozen block leads by +21.3, +23.3, +20.3 pp. The multiplicative output gate is the ablated component whose removal costs most, but under independent noise this cost is resolved only at -6 dB, and without its gate the fully frozen block still exceeds S4D at milder noise: the gate’s contribution

grows with noise depth rather than carrying the advantage throughout. An additive branch of identical size, placed where the gate was, could not be distinguished from removing it; added to S4D, a gate recovers part of the gap, not more than an additive branch of the same size on one direction and somewhat more on the other. On Paderborn, the gate-removal cost replicates when bearings are shared between training and evaluation and nothing replicates on disjoint bearings, where every architecture transfers poorly. The evidence rests on one block, one shift pair per dataset, five seeds and final-epoch reporting under a fixed budget. We offer the three-step chain—freeze, remove, replace—as a protocol for turning architecture comparisons into component-level tests, with two cautions from this study: interventions must be matched in width and capacity, and the noise streams of training and evaluation must be independent.

CRedit authorship contribution statement

Yi Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used LLM-based agents to execute and cross-check pre-specified analysis and adjudication passes as described in Appendix C, and to assist with manuscript drafting and figure generation. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. All numeric thresholds, adjudication rulings, and scientific interpretations reported in this paper were fixed and made by the human authors, as detailed in Appendix C.

906 Data availability

907 The code, pre-specification artifacts, and adjudication trail for this study
908 are available at <https://github.com/jeff-1980/bm3-jizi>. The repository
909 includes all follow-up grids (585 cells with per-epoch curves, including the
910 150 independent-noise cells of Section 4.1), their pre-specifications, drivers,
911 construction checks and decision memos and the harness’s own imported
912 modules, so the training entry point runs from a fresh checkout.

913 Appendix A. Fairness protocol

914 *Appendix A.1. Evaluation-set fingerprint*

915 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
916 256 hash of the raw held-out test-label byte array, computed once per run
917 and asserted identical across every arm/condition/seed cell within a grid.
918 It fingerprints *that the same held-out label vector was scored every time*, not
919 model weights or predictions. It is also the split on which the best-epoch rule
920 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
921 five grids is:

922 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

923 This value is recomputed independently at adjudication time (not merely
924 copied from the run log) and cross-checked row-by-row against every grid’s
925 fairness report before any comparison across arms is treated as valid.

926 *Appendix A.2. Per-arm parameter counts*

927 Table A.14 lists every arm’s parameter count, each obtained directly via
928 `sum(p.numel() for p in model.parameters())` on the instantiated model
929 (not estimated), cross-verified against an independently maintained archi-
930 tecture map.

931 *Appendix A.3. Five-batch pre-specification timeline*

932 Table A.15 lists, for each of the five pre-specified grids, the artifact that
933 fixed its adjudication thresholds, that artifact’s creation/modification date,
934 and the corresponding grid’s run-directory start time. In every case the
935 prereg artifact predates the grid it governs.

936 Two honest caveats. First, the baseline batch predates the orchestration
937 infrastructure used for the other four batches by five days; its prereg trail

Table A.14: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

Table A.15: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (mtime 18:26:54)	2026-07-02	2026-07-02 18:27
Component ablation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

938 rests on the task card’s creation timestamp alone, not on the same automat-
939 ed executor/reviewer machinery documented in [Appendix C](#), and we state
940 this rather than implying a uniform process across all five batches. Second,
941 for the component-ablation grid, two additional JSON files bearing “pre-
942 reg” in their name exist on disk (`secondary_prereg_20260704-0708` and
943 `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start
944 time; both are explicitly marked void in the corresponding decision artifact’s
945 `void_preregs` field, and neither was used to adjudicate any result in this pa-
946 per. They are the on-disk residue of the first post-hoc criterion substitution
947 described in [Appendix C](#), retained here (rather than deleted) as part of the
948 released adjudication trail.

949 **Appendix B. Confusion matrices and a repeat-run check**

950 *Appendix B.1. Full confusion-matrix set*

951 Table [B.16](#) gives the complete confusion-matrix set underlying the per-
952 class analysis of [Section 4.11](#): 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),
953 each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These
954 counts were dumped per-sample and re-verified against the reported recal-
955 l/precision/F1 metrics cell-by-cell before use.

Table B.16: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true $\rightarrow$ predicted counts.

Arm	Condition	OR $\rightarrow$ OR	OR $\rightarrow$ IR	IR $\rightarrow$ OR	IR $\rightarrow$ IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	-6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	-6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
	-6 dB	13,267	4,253	3,244	5,476
frozen-gate	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
	-6 dB	13,200	4,320	5,469	3,251

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
S4D	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

956 *Appendix B.2. Repeat-run check*

957 Because two independent process runs of an identical configuration are
958 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
959 ductions in the selective-scan and normalisation ops), we measured a process-
960 level noise floor directly: the same arm, condition, and seed (`bm3_frozen`,
961 clean, seed 0, 2 epochs) was run twice in separate processes with identical
962 evaluation fingerprints and identical parameter counts confirmed. The two
963 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

964 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
965 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
966 run, and therefore neither an estimate nor a bound of process variability for
967 full-length runs. It is an order of magnitude below the seed-level variabil-
968 ity observed within the deep-noise band across the curve tables underlying
969 Figures 2 and 4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–
970 9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4
971 are judged against seed-level variability, not against this single repeat.

972 **Appendix C. Automation disclosure and adjudication gates**

973 *Appendix C.1. Automation disclosure*

974 The mechanical stages of the analysis pipeline—applying pre-specified nu-
975 meric thresholds to aggregated results, re-computing aggregates, re-verifying
976 evaluation fingerprints, and checking each decision against its pre-specified
977 rule—were carried out by LLM-based coding agents under an orchestration
978 harness, with every step auditable in the released trail. Manuscript drafting
979 and figure generation were also assisted by such agents (main text, Declara-
980 tion of Generative AI).

981 *Appendix C.2. Incidents*

982 Three incidents occurred in the original grids, each caught before an er-
983 roneous artefact entered a reported number: (1) on the frozen-selectivity
984 grid an automated decision substituted an unspecified criterion for the pre-
985 specified 10 pp threshold; re-applying the original rule gives the reported out-
986 come, “not the primary mechanism”; (2) on the component-ablation grid an
987 automated pass substituted an interval heuristic for the pre-specified thresh-
988 old, and post-hoc rule files were generated; the true pre-specification predated
989 the grid and was re-applied, and the post-hoc files are retained and marked
990 void; (3) on the graft grid a construction stage was falsely marked as passed;
991 the grid was re-executed from scratch under a corrected task card. Devia-
992 tions in the later follow-up grids (for example a bootstrap made deterministic
993 after first use, and rounding corrections) are recorded in the repository’s de-
994 cision memos. Full descriptions, with dates and artefact names, are given in
995 Supplementary Material S1.

996 *Appendix C.3. Human adjudication gates*

997 All numeric adjudication thresholds, every ruling on an incident, the de-
998 cision to report the incidents, and every interpretive claim in Section 5 were
999 fixed or made by the human authors. No agent could overwrite a protected
1000 decision artefact, alter a pre-specified threshold or mark its own output as
1001 final.

1002 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

1003 In line with Elsevier’s policy, AI tools are not listed as authors and the
1004 human authors take full responsibility for the content. We extend the dis-
1005 closure beyond the writing process to the adjudication pipeline because that
1006 is where automation could affect the reported conclusions.

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